UNIVERSITY OF KELANIYA / SENG 34213

FOREVER HOTEL MANAGEMENT SYSTEM

# OWASP Compliance Evidence

OWASP Top 10 (2021) Control Review | Six Subsystems

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Reporting date: 9 November 2026. Evidence is identified by subsystem repository, branch, commit SHA, and GitHub Actions run date; comparisons across separate repository commits do not constitute a single deployed hotel-wide security certification.

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# 1. Security evidence overview

Security controls across all six Forever Hotel subsystems and companion shared services (api-gateway, websocket-service, infra) are evaluated against the OWASP Top 10 (2021) categories A01 through A10 as of 9 November 2026. Passing unit, integration, and E2E security assertions demonstrate application-level controls within their tested boundaries; they do not substitute for a live staging TLS/WAF audit, independent penetration test, or production compliance certification.

## Dependency-audit results and their scope (9 November 2026)

### Hotel Website (forever-hotel/hotel-website)

In the committed HW OWASP checklist (commit 63673ba704, documents/owasp-checklist.pdf), raw full-tree npm audit --json recorded 4 backend advisories (3 high: @nestjs/platform-express, body-parser, multer; 1 moderate: fast-uri), 1 critical frontend advisory (next, fix available in 16.4.3), and 2 in-repo gateway advisories (1 moderate: fast-uri; 1 low: fastify). On develop (5e3cd6bf32, 9 November 2026), all backend, frontend, container, and staging CI checks pass, while full-tree dependency upgrades remain tracked as an explicit release gate.

### Manager Dashboard & Shared Services (manager-dashboard, api-gateway, websocket-service)

REMEDIATED IN CI: The previous Manager Dashboard backend high-severity audit failure (body-parser, multer, qs) was resolved via package.json overrides on branch development (commit 8b2c63f459, Runs 42480400290 and 42504148805). Both backend (npm audit --audit-level=high) and frontend (npm run audit:ci) now pass with 0 high/critical vulnerabilities. In addition, the dedicated shared repositories forever-hotel/api-gateway (commit f12aad6cb4, Run 42504067779) and forever-hotel/websocket-service (commit f12ef41ba9, Run 42504249365) both pass npm audit --audit-level=high with 0 high/critical vulnerabilities.

### Front Desk System (forever-hotel/front-desk)

REMEDIATED IN CI: The previous high-severity undici development dependency finding was remediated by upgrading undici to 8.12.2 (commit 1065617 on feature/84-fds-final-qa-evidence and merged into develop at 4b5ef57d4c, Runs 42506217082 and 42506217056). Backend high-threshold audit, frontend production-only audit, and full frontend npm audit all return 0 vulnerabilities.

### Food Ordering and Service Requests (forever-hotel/food-ordering)

PASSING IN CI: On develop (commit 0d80cf5c66, Run 42490527447) and QA snapshot 9ac1e6f (Run 41833172660), the dedicated Dependency Security Scan CI job, Oxlint (240 backend files, 0 errors/warnings), and ESLint 9 (frontend, 0 errors/warnings) all pass with 0 high/critical vulnerabilities.

### Kitchen Management System (forever-hotel/kitchen-management)

PASSING IN CI: On develop (commit 2bfa14b553, Run 42502180570), both backend and frontend npm run security:audit (npm audit --omit=dev --audit-level=high) pass with 0 vulnerabilities across production dependencies alongside 665 passing automated tests.

### Worker Management System (forever-hotel/worker-management)

PASSING IN CI: On develop (commit 1a33059c00, Runs 42504698971 and 42504970394), both backend and frontend production high-threshold dependency audits pass with 0 vulnerabilities alongside 649 passing automated tests and Docker image verification.

## Control coverage

All 60 subsystem/category combinations (six subsystems multiplied by ten OWASP Top 10 categories) are evaluated below, preserving verified source/test evidence and remaining staging or operational verification actions.

# 2. Hotel Website (HW)

Repository: forever-hotel/hotel-website | Branch: develop (merged PR #99 from pre-develop 440761c34a; QA package: 63673ba704b4ae645a8f067e246309b6fb0f03d0) | Tested revision: 5e3cd6bf323bc005e9efb86b5e09ba0e3015e727 | CI date: 9 November 2026

Evidence: https://github.com/forever-hotel/hotel-website/actions/runs/42504096480, https://github.com/forever-hotel/hotel-website/actions/runs/42504096452, https://github.com/forever-hotel/hotel-website/actions/runs/42500427127, documents/owasp-checklist.pdf (63673ba704)

Dependency position: Raw full-tree npm audit --json in documents/owasp-checklist.pdf (63673ba704) recorded 3 high + 1 moderate backend advisories (@nestjs/platform-express, body-parser, multer, fast-uri) and 1 critical frontend advisory (next, fix in 16.4.3); standalone forever-hotel/api-gateway (f12aad6cb4) passes npm audit --audit-level=high with 0 high/critical findings.

## A01 Broken Access Control

Broken Access Control — PARTIALLY EVIDENCED: JWT guest guards and guest-scoped repositories enforce strict booking ownership; cross-guest read, modification, cancellation, and payment recovery attempts are verified in disposable PostgreSQL HTTP integration tests (returning 404 anti-enumeration responses rather than 403), and frontend returnTo validation blocks open redirects. Evidence: backend/src/common/guards/jwt-auth.guard.ts; backend/test/integration/http-consumer-contracts.integration-spec.ts (passing in Run 42504096480); frontend/tests/auth.test.mjs.

Required follow-up: Obtain reviewer sign-off on the 401/404 anti-enumeration contract and verify access controls on deployed staging.

## A02 Cryptographic Failures

Cryptographic Failures — PARTIALLY EVIDENCED: Guest passwords use bcrypt cost 12; JWT tokens use timing-safe HMAC verification with 24h expiry; session cookies use HttpOnly and production Secure attributes; PayHere webhook notifications verify merchant HMAC signatures before payment reconciliation. Evidence: backend/src/guests/password-hasher.service.spec.ts; backend/src/common/utils/jwt-token.service.spec.ts; backend/src/payments/payhere-provider.spec.ts; frontend/src/features/auth/actions.ts.

Required follow-up: Verify staging HTTPS/TLS certificate configuration and secret rotation procedures.

## A03 Injection

Injection — PARTIALLY EVIDENCED: SQL repositories use parameterized queries exclusively; NestJS ValidationPipe enforces strict DTO whitelisting and input bounds; React escapes rendered text; room catalogue and image utilities reject unsafe origins and malformed records. Evidence: backend/src/database/repositories/booking.repository.ts; backend/test/integration/http-consumer-contracts.integration-spec.ts; frontend/tests/rooms.test.mjs.

Required follow-up: Execute automated DAST/fuzzing scans against deployed staging routes.

## A04 Insecure Design

Insecure Design — PARTIALLY EVIDENCED: Signed change previews bind guest identity, booking state, and revision; competing final-room reservations use sorted row locks verified by a two-guest concurrent PostgreSQL race test (booking-race.integration-spec.ts passing in Run 42504096480); client redirect parameters cannot confirm unpaid bookings (TC-HW-11-02, TC-SEC-03). Evidence: backend/src/bookings/booking-management.service.spec.ts; backend/test/integration/booking-race.integration-spec.ts; frontend/tests/payment.test.mjs; deploy/staging/test/smoke.test.mjs.

Required follow-up: Complete live MAD promotion redemption and live PayHere sandbox callback verification.

## A05 Security Misconfiguration

Security Misconfiguration — PARTIALLY EVIDENCED: Production configuration disables Swagger and rejects DB_LOGGING=true (TC-SEC-02) and schema synchronization; AllExceptionsFilter returns generic client errors without leaking stack traces, SQL, or secrets (TC-SEC-01); container CI verifies 503 readiness behavior during PostgreSQL outage. Evidence: backend/src/config/env.validation.spec.ts; backend/src/common/filters/all-exceptions.filter.spec.ts; .github/workflows/container-checks.yml (Run 42504096444).

Required follow-up: Verify deployed reverse-proxy security headers (CSP, HSTS), container hardening, and vault secret injection.

## A06 Vulnerable and Outdated Components

Vulnerable and Outdated Components — OPEN FINDINGS (TRACKED): Raw npm audit --json in documents/owasp-checklist.pdf (63673ba704) identified 3 high backend transitive package findings (@nestjs/platform-express, body-parser, multer), 1 moderate backend finding (fast-uri), and 1 critical frontend advisory (next, fix available in 16.4.3). Evidence: backend/package-lock.json; frontend/package-lock.json; documents/owasp-checklist.pdf.

Required follow-up: Apply compatible Next.js and NestJS/Express transitive dependency upgrades (matching the overrides applied in MAD), rerun full test and browser suites, and enforce a blocking CI audit gate.

## A07 Identification and Authentication Failures

Identification and Authentication Failures — PARTIALLY EVIDENCED: Generic unauthorized responses prevent email enumeration (TC-HW-016); expired/tampered JWTs are rejected; contact enquiry endpoint enforces per-IP rate limiting (TC-HW-16-02); API Gateway enforces per-process login rate limiting. Evidence: backend/test/integration/guest-auth.integration-spec.ts; backend/src/contact/contact-rate-limit.guard.spec.ts; api-gateway/src/app.ts.

Required follow-up: Implement account-based lockout after repeated failed logins and evaluate token revocation/distributed throttling.

## A08 Software and Data Integrity Failures

Software and Data Integrity Failures — PARTIALLY EVIDENCED: CI uses npm ci with committed lockfiles; staging release tooling enforces digest-pinned immutable images and automatic rollback to the previous image pair on health/smoke failure (TC-DEPLOY-01, 22 passing tests in Run 42500427127); PayHere webhook verification rejects tampered amounts or signatures. Evidence: deploy/staging/test/release.test.mjs; deploy/staging/test/smoke.test.mjs; backend/src/payments/payhere-provider.spec.ts.

Required follow-up: Verify live staging rollback execution and signed container image provenance.

## A09 Security Logging and Monitoring Failures

Security Logging and Monitoring Failures — PARTIALLY EVIDENCED: Secret, card, and SQL canary regression tests verify that AllExceptionsFilter never logs sensitive payloads (TC-SEC-01); payment and email providers emit structured failure events without raw provider payloads. Evidence: backend/src/common/filters/all-exceptions.filter.spec.ts; backend/src/payments/payhere-payments.service.spec.ts.

Required follow-up: Connect container logs to centralized log retention and configure security alerting on authentication/payment anomalies.

## A10 Server-Side Request Forgery

Server-Side Request Forgery — PARTIALLY EVIDENCED: Room image helpers enforce an explicit HTTPS host allowlist and reject credentials or unsafe paths (TC-HW-03-01); payment return/redirect targets are restricted; gateway upstream origins are server-configured. Evidence: frontend/src/config/room-images.ts; frontend/tests/rooms.test.mjs; frontend/tests/payment.test.mjs.

Required follow-up: Upgrade Next.js to remediate Image Optimization advisories and enforce egress network policies in staging.

# 3. Manager Dashboard (MAD)

Repository: forever-hotel/manager-dashboard (+ forever-hotel/api-gateway f12aad6cb4, forever-hotel/websocket-service f12ef41ba9, forever-hotel/infra e3e17fcaac) | Branch: development | Tested revision: 8b2c63f459a93c217eb7cb502a2500e80181aa6e | CI date: 9 November 2026

Evidence: https://github.com/forever-hotel/manager-dashboard/actions/runs/42480400290, https://github.com/forever-hotel/manager-dashboard/actions/runs/42504148805, https://github.com/forever-hotel/api-gateway/actions/runs/42504067779, https://github.com/forever-hotel/websocket-service/actions/runs/42504249365

Dependency position: REMEDIATED IN CI — Backend npm audit --audit-level=high and frontend npm run audit:ci both passed with 0 high/critical vulnerabilities in Runs 42480400290 and 42504148805 after pinning transitive overrides (multer, body-parser, qs) in backend/package.json. Extracted shared repositories forever-hotel/api-gateway and forever-hotel/websocket-service also passed npm audit --audit-level=high with 0 high/critical vulnerabilities.

## A01 Broken Access Control

Broken Access Control | PARTIALLY VERIFIED: Manager JWT/role guards, protected frontend route navigation, read-only operational views, and API Gateway route-level RBAC checks are verified across 534 MAD backend unit tests, 203 PostgreSQL/RabbitMQ integration tests, 8 Playwright E2E tests, and 39 API Gateway tests. Evidence: backend/src/common/security-boundaries.spec.ts; frontend/test/protection.test.ts; forever-hotel/api-gateway Run 42504067779.

Required follow-up: Execute full multi-role penetration testing across deployed gateway and subsystem endpoints.

## A02 Cryptographic Failures

Cryptographic Failures | PARTIALLY VERIFIED: JWT signing/verification, bcrypt password hashing (cost >=12), and encrypted staff credential delivery are unit- and integration-tested in MAD and verified across central auth / WebSocket system checks (npm run test:system). Evidence: backend/src/auth/token.service.spec.ts; backend/src/staff/staff.spec.ts; Run 42480400290.

Required follow-up: Verify production TLS termination, vault secret management, and key rotation in deployed staging.

## A03 Injection

Injection | PARTIALLY VERIFIED: Strict DTO validation, parameterized PostgreSQL queries, and CSV formula injection sanitization are verified in unit and PostgreSQL integration suites. Evidence: backend/src/promotions/promotions.spec.ts; backend/src/reports/reports.spec.ts.

Required follow-up: Perform automated DAST fuzzing against reporting, filtering, and search endpoints.

## A04 Insecure Design

Insecure Design | PARTIALLY VERIFIED: Optimistic concurrency version checks, login lockout rules, atomic promotion and complaint workflows, and transactional outbox publication retries are verified in unit and PostgreSQL/RabbitMQ integration suites. Evidence: backend/test/promotions.e2e-spec.ts; backend/test/complaints.e2e-spec.ts.

Required follow-up: Obtain formal cross-subsystem threat-model sign-off.

## A05 Security Misconfiguration

Security Misconfiguration | PARTIALLY VERIFIED: Frontend security response headers, strict environment validation, and infrastructure Compose/release configuration checks (19 passing tests in forever-hotel/infra Run 42501357369) are verified in CI. Evidence: frontend/test/security-headers.test.ts; backend/src/config/environment.spec.ts; forever-hotel/infra Run 42501357369.

Required follow-up: Inspect live staging reverse-proxy headers and container runtime privileges.

## A06 Vulnerable and Outdated Components

Vulnerable and Outdated Components | REMEDIATED IN CI (PASS): Transitive high-severity advisories in body-parser, multer, and qs were remediated via package.json overrides; backend npm audit --audit-level=high, frontend npm run audit:ci, api-gateway npm audit --audit-level=high, and websocket-service npm audit --audit-level=high all passed in 9 November 2026 CI. Evidence: MAD Runs 42480400290 & 42504148805; API Gateway Run 42504067779; WebSocket Service Run 42504249365.

Required follow-up: Maintain continuous Dependabot and lockfile auditing up to final release tagging.

## A07 Identification and Authentication Failures

Identification and Authentication Failures | PARTIALLY VERIFIED: Active-account checks, login lockout, JWT validation, mandatory first-password change, central auth system integration (npm run test:system), and WebSocket handshake authentication (36 passing tests in forever-hotel/websocket-service) are verified in CI. Evidence: backend/src/auth/auth.service.spec.ts; frontend/test/protection.test.ts; forever-hotel/websocket-service Run 42504249365.

Required follow-up: Verify cross-subsystem staff login and session revocation on deployed staging.

## A08 Software and Data Integrity Failures

Software and Data Integrity Failures | PARTIALLY VERIFIED: CI enforces npm ci with locked dependencies, changed-code coverage gates (check-changed-coverage.test.mjs: 5 passed), build config verification (verify-build-config.test.mjs), and outbox publication confirmation/retry logic. Evidence: .github/workflows/ci.yml; backend/src/promotions/promotion-publication.spec.ts.

Required follow-up: Establish signed container image provenance for production deployment.

## A09 Security Logging and Monitoring Failures

Security Logging and Monitoring Failures | PARTIALLY VERIFIED: Staff credential reset auditing, profile change auditing, log redaction, and safe error responses are verified in unit and integration tests. Evidence: backend/src/staff/staff-reset.spec.ts; backend/src/staff/staff-profiles.spec.ts.

Required follow-up: Verify centralized 12-month log retention and real-time alerting SLAs in staging.

## A10 Server-Side Request Forgery

Server-Side Request Forgery | PARTIALLY VERIFIED: API Gateway routing and WebSocket upstream connections are bound to server-configured service origins validated in forever-hotel/api-gateway and forever-hotel/infra tests; no user-supplied outbound URL fetching is exposed. Evidence: forever-hotel/api-gateway Run 42504067779; forever-hotel/infra Run 42501357369.

Required follow-up: Execute targeted SSRF and egress-filtering verification in the deployed container network.

# 4. Front Desk System (FDS)

Repository: forever-hotel/front-desk | Branch: develop (QA evidence branch: feature/84-fds-final-qa-evidence) | Tested revision: 4b5ef57d4c4ce03c9ab2fe78f42b94eff96838cd (QA snapshot: e590326 / 1065617 / a6050dd) | CI date: 9 November 2026

Evidence: https://github.com/forever-hotel/front-desk/actions/runs/42506217082, https://github.com/forever-hotel/front-desk/actions/runs/42506217056, https://github.com/forever-hotel/front-desk/actions/runs/40530366600, https://github.com/forever-hotel/front-desk/actions/runs/40534533200

Dependency position: REMEDIATED IN CI — undici dev dependency upgraded to 8.12.2 (commit 1065617 and develop 4b5ef57d4c). Backend CI audit, frontend production-only audit, and full frontend npm audit all return 0 vulnerabilities.

## A01 Broken Access Control

Broken Access Control — PARTIALLY EVIDENCED: JWT authentication, read/write RBAC enforcement, 401/403 HTTP E2E checks (worker role denied booking reads in TC-FDS-041), and local authenticated FDS-to-WKMS manual assignment permission checks (W04-B: 6 PASS) are verified across 143 HTTP E2E tests and 531 unit tests on develop (4b5ef57d4c). Source: backend/test/bookings-rbac.e2e-spec.ts; backend/test/check-in-rbac.e2e-spec.ts; backend/test/rooms-rbac.e2e-spec.ts; backend/test/audit-rbac.e2e-spec.ts.

Follow-up: Complete exhaustive endpoint-by-role negative matrix and public staging gateway review.

## A02 Cryptographic Failures

Cryptographic Failures — NOT FULLY VERIFIED: JWT signature, expiry, and issuer verification are unit- and E2E-tested; checkout DTO validation blocks raw card numbers/CVV (TC-FDS-042). Central identity password hashing (bcrypt >=12) and staging HTTPS/TLS termination remain external to FDS CI. Source: backend/src/security/auth/jwt-principal.resolver.spec.ts; backend/test/folio-checkout-rbac.e2e-spec.ts.

Follow-up: Verify central auth credential storage and staging HTTPS/TLS certificates.

## A03 Injection

Injection — PARTIALLY EVIDENCED: PostgreSQL repositories use parameterized queries exclusively (56 passing PostgreSQL integration tests on 4b5ef57d4c); DTO validation and plain-text sanitization strip unsafe input. Source: backend/src/reservations/repositories/postgres-booking.repository.ts; backend/src/security/sanitization/sanitize-plain-text.spec.ts.

Follow-up: Execute route-level injection and fuzzing scans against deployed staging.

## A04 Insecure Design

Insecure Design — PARTIALLY EVIDENCED: Atomic PostgreSQL transactions with rollback on payment failure (TC-FDS-008), room conflict rejection (TC-FDS-011, TC-FDS-022), and database-enforced append-only audit triggers rejecting UPDATE/DELETE (TC-FDS-031, TC-FDS-032) are verified in CI and local Plan 02 (35 PASS). Source: backend/test/database/postgres-checkout.repository.integration-spec.ts; backend/test/database/audit-append-only.integration-spec.ts.

Follow-up: Obtain formal cross-service threat-model sign-off.

## A05 Security Misconfiguration

Security Misconfiguration — PARTIALLY EVIDENCED: Joi startup validation enforces staging/production secrets, blocks wildcard CORS origins, and blocks DB_SYNCHRONIZE; Neon test branch schema & audit triggers (9/9 tables, 3/3 triggers) were verified in DDP-195. Source: backend/src/config/env.validation.spec.ts; evidence/ddp195_verification.md.

Follow-up: Reconcile Neon TypeORM migration history table and verify deployed staging headers.

## A06 Vulnerable and Outdated Components

Vulnerable and Outdated Components — REMEDIATED IN CI (PASS): undici upgraded to 8.12.2 in commit 1065617 and merged into develop (4b5ef57d4c). Full frontend audit, production-only audit, and backend audit all report 0 vulnerabilities in Runs 40534533200, 42506217056, and 42506217082. Source: frontend/package-lock.json; backend/package-lock.json.

Follow-up: Confirm GitHub Dependabot alert closure on default branch.

## A07 Identification and Authentication Failures

Identification and Authentication Failures — PARTIALLY EVIDENCED: Invalid, expired, and wrong-issuer JWTs are rejected on both HTTP routes and Socket.IO handshakes (TC-FDS-043); Phase D browser verification (8 PASS) confirmed realtime auth and reconnect behavior. Source: backend/src/security/auth/jwt-principal.resolver.spec.ts; backend/test/realtime.e2e-spec.ts.

Follow-up: Verify end-to-end central auth receptionist login, lockout, and session rotation.

## A08 Software and Data Integrity Failures

Software and Data Integrity Failures — PARTIALLY EVIDENCED: CI runs npm ci with lockfiles, verifies multi-stage Docker builds (Run 42506217053), and enforces append-only audit log triggers in PostgreSQL; Phase C verified RabbitMQ broker restart and consumer recovery (5 PASS). Source: backend/test/database/audit-append-only.integration-spec.ts; .github/workflows/docker-image.yml.

Follow-up: Verify signed container artifact provenance in release deployment.

## A09 Security Logging and Monitoring Failures

Security Logging and Monitoring Failures — PARTIALLY EVIDENCED: Check-in, checkout, room change, maintenance, and manual WKMS assignment operations write immutable audit records; LogRedactor strips PII and secrets from logs. Source: backend/src/security/redaction/log-redactor.spec.ts; backend/test/audit-rbac.e2e-spec.ts.

Follow-up: Configure centralized log shipping, retention, and alerting on repeated 401/403 events.

## A10 Server-Side Request Forgery

Server-Side Request Forgery — PARTIALLY EVIDENCED: Outbound HTTP calls to WKMS use strictly configured service base URLs (wkms-http.client.spec.ts) and were exercised in local W04-B authenticated assignment verification (6 PASS). Source: backend/src/integrations/wkms/wkms-http.client.spec.ts.

Follow-up: Enforce outbound network allowlists and SSRF negative tests in deployed staging.

# 5. Food Ordering and Service Requests (FOSS)

Repository: forever-hotel/food-ordering | Branch: develop | Tested revision: 0d80cf5c66b566580927cb8fae1f7d549a15f03f (QA package commit: 9ac1e6f3df9f93ee76be6db479dc8c4bf6529328) | CI date: 9 November 2026

Evidence: https://github.com/forever-hotel/food-ordering/actions/runs/42490527447, https://github.com/forever-hotel/food-ordering/actions/runs/41833172660, documents/security/owasp-checklist.pdf & documents/security/owasp-notes.md (commit 9ac1e6f)

Dependency position: PASSING IN CI — Dedicated Dependency Security Scan job in Run 42490527447 (9 November 2026), Oxlint (240 backend files, 0 errors/warnings), and ESLint 9 (frontend, 0 errors/warnings) all passed with 0 high/critical vulnerabilities.

## A01 Broken Access Control

Broken Access Control — PARTIALLY EVIDENCED: Guest endpoints require a valid signed Bearer JWT; RoomOwnershipGuard verifies that token roomNumber and bookingId match target resources, rejecting cross-room (Room 101 -> Room 102) and cross-booking IDOR attempts with 403 Forbidden (TC-FOSS-009, TC-FOSS-010); FDS checkout events immediately revoke active sessions (TC-FOSS-007). Source: backend/src/common/auth/room-ownership.guard.ts; backend/test/room-scoped-auth.e2e-spec.ts; backend/test/fds-checkout-session-revocation.e2e-spec.ts; frontend/tests/fui-02-room-isolation.test.ts.

Required follow-up: Coordinate with API Gateway to enforce IP rate limiting on /foss/guest-sessions/qr and execute multi-tenant penetration tests.

## A02 Cryptographic Failures

Cryptographic Failures — PARTIALLY EVIDENCED: QR activation tokens are signed with HMAC-SHA256 (QR_HMAC_SECRET, TC-FOSS-002); guest JWTs are transmitted strictly via Authorization: Bearer headers; fetch-based SSE streaming avoids exposing tokens in EventSource URLs (TC-FOSS-041); Neon PostgreSQL enforces TLS 1.3. Source: backend/src/modules/guest-sessions/services/qr-activation.service.spec.ts; backend/test/qr-activation-lifecycle.e2e-spec.ts; frontend/tests/realtime-client-security.test.ts.

Required follow-up: Audit production API Gateway TLS cipher suites and confirm vault secret rotation.

## A03 Injection

Injection — PARTIALLY EVIDENCED: TypeORM repositories use parameterized SQL queries exclusively; global NestJS ValidationPipe (whitelist: true, forbidNonWhitelisted: true) strips undeclared fields; free-text inputs are bounded (allergy notes <=300 chars in TC-FOSS-022, complaint and service-request descriptions <=500 chars); sanitizePlainText strips HTML/script tags. Source: backend/src/modules/food-orders/dto/create-order.dto.ts; backend/test/api-validation-and-errors.e2e-spec.ts; frontend/tests/pricing-and-quantity.test.ts.

Required follow-up: Run automated OWASP ZAP baseline API scan against containerized backend routes.

## A04 Insecure Design

Insecure Design — PARTIALLY EVIDENCED: High-entropy Idempotency-Key headers prevent duplicate orders and double-billing (TC-FOSS-025); 10-minute cancellation cutoff and kitchen PREPARING lock-in are enforced on backend (TC-FOSS-030..032); server recalculates order totals from authoritative catalog prices in integer minor units (TC-FOSS-023, TC-FOSS-024); cart quantity is capped at 20 per line item (TC-FOSS-019). Source: backend/src/modules/food-orders/food-orders.service.spec.ts; backend/test/food-orders.e2e-spec.ts; frontend/tests/cart.test.ts.

Required follow-up: Complete architectural review for PostgreSQL transactional outbox (docs/change-requests/CR-outbox-foss.md).

## A05 Security Misconfiguration

Security Misconfiguration — PARTIALLY EVIDENCED: class-validator bootstrap validation (env.validation.ts) rejects missing config; HealthService and assertApprovedDatabaseIdentity block test suites from running against development/production Neon branches and enforce synchronize: false; frontend blocks DATA_MODE=demo in production; HttpExceptionFilter prevents stack trace disclosure. Source: backend/src/config/env.validation.ts; backend/src/health/health.service.spec.ts; backend/src/database/guards/assert-approved-database-identity.spec.ts; backend/src/common/filters/http-exception.filter.ts.

Required follow-up: Verify gateway security headers (CSP, HSTS) and non-root container execution in staging.

## A06 Vulnerable and Outdated Components

Vulnerable and Outdated Components — PARTIALLY EVIDENCED (CI SCAN PASS): Dependency Security Scan job in GitHub Actions Run 42490527447 passed with 0 high/critical vulnerabilities; Oxlint (240 backend files) and ESLint 9 (frontend) pass with 0 errors/warnings; lockfiles are strictly pinned. Source: backend/package.json; frontend/package.json; GitHub Actions Run 42490527447.

Required follow-up: Maintain continuous Dependabot monitoring across transitive dev dependencies.

## A07 Identification and Authentication Failures

Identification and Authentication Failures — PARTIALLY EVIDENCED: QR slip activation validates booking reference, room number, and high-entropy access code (TC-FOSS-001, TC-FOSS-002); sessions enforce a 24h TTL (SESSION_TTL_HOURS=24, TC-FOSS-006) and immediate revocation on FDS checkout (TC-FOSS-007) or SSE SESSION_REVOKED events (TC-FOSS-044); frontend partitions demo and API tokens cleanly (TC-FOSS-011, TC-FOSS-012). Source: backend/src/modules/guest-sessions/guest-sessions.service.ts; backend/test/guest-sessions.e2e-spec.ts; frontend/tests/auth-lifecycle-isolation.test.ts.

Required follow-up: Verify gateway brute-force rate limiting on /foss/guest-sessions/qr.

## A08 Software and Data Integrity Failures

Software and Data Integrity Failures — PARTIALLY EVIDENCED: RabbitMQ inter-service messages require HMAC-SHA256 SignedEventEnvelope verification (TC-FOSS-027, TC-FOSS-045); event ID deduplication discards replayed messages (TC-FOSS-046); malformed/poisoned payloads route to forever.hotel.dlq (TC-FOSS-047); TypeORM enforces synchronize: false. Source: backend/src/integrations/contracts/signed-event-envelope.spec.ts; backend/src/messaging/rabbitmq-consumer.service.spec.ts; backend/test/rabbitmq-live-integration.e2e-spec.ts.

Required follow-up: Enforce RabbitMQ TLS and per-service vhost/credential isolation in production.

## A09 Security Logging and Monitoring Failures

Security Logging and Monitoring Failures — PARTIALLY EVIDENCED: CorrelationMiddleware propagates X-Correlation-ID across requests; LogSanitizerUtil masks Bearer tokens, passwords, and guest PII to [REDACTED] before logging; security lifecycle events (QR activation, revocation, order cancellation) emit structured logs. Source: backend/src/common/middleware/correlation.middleware.spec.ts; backend/src/common/logging/log-sanitizer.util.spec.ts.

Required follow-up: Integrate container stdout streams with centralized hotel SIEM/Loki log aggregation and alerting.

## A10 Server-Side Request Forgery

Server-Side Request Forgery — PARTIALLY EVIDENCED: Outbound HTTP adapters for FDS, KMS, WKMS, and MAD use strictly preconfigured environment base URLs and hardcoded path templates without accepting user-supplied URLs or following redirects. Source: backend/src/integrations/internal-services/booking/adapters/real-booking.adapter.ts; backend/src/integrations/internal-services/kms/adapters/real-kms.adapter.ts; backend/src/integrations/internal-services/wkms/adapters/real-wkms.adapter.ts; backend/src/integrations/internal-services/mad/adapters/real-mad-complaint.adapter.ts.

Required follow-up: Enforce container egress policies blocking access to cloud metadata endpoints (169.254.169.254).

# 6. Kitchen Management System (KMS)

Repository: forever-hotel/kitchen-management | Branch: develop | Tested revision: 2bfa14b5535c5c065f49dbdc2b7c62c4c8105d15 (PR #101) | CI date: 9 November 2026

Evidence: https://github.com/forever-hotel/kitchen-management/actions/runs/42502180570

Dependency position: PASSING IN CI — Both backend and frontend npm run security:audit (npm audit --omit=dev --audit-level=high) passed with 0 vulnerabilities in Run 42502180570 alongside 665 passing automated unit, E2E, and frontend tests.

## A01 Broken Access Control

Broken Access Control — PARTIALLY EVIDENCED: Between 10 October and 9 November 2026 (commit 2bfa14b553, Run 42502180570), KMS expanded from 2 scaffold E2E suites to 10 passing E2E suites (58 passing E2E tests) and 54 backend unit suites (234 passing unit tests) covering kitchen order creation/reading/state transitions and menu item/category write operations. Cross-service gateway RBAC is enforced in forever-hotel/api-gateway (39 passing tests).

Remaining verification: Execute end-to-end negative RBAC penetration tests across all kitchen staff and manager roles on deployed staging.

## A02 Cryptographic Failures

Cryptographic Failures — PARTIALLY EVIDENCED: Database configuration tests verify managed PostgreSQL SSL and certificate verification settings (database.e2e-spec.ts); central JWT verification is handled at the API Gateway / auth boundary.

Remaining verification: Verify staging TLS certificates, broker transport encryption, and secret rotation.

## A03 Injection

Injection — PARTIALLY EVIDENCED: Joi environment validation, NestJS DTO validation across menu and order write endpoints (menu-item-write.e2e-spec.ts, menu-category-write.e2e-spec.ts, orders-create.e2e-spec.ts), allergy note safety validation (allergy-safety.e2e-spec.ts), and parameterized TypeORM repository queries are verified in Run 42502180570.

Remaining verification: Execute hostile SQL/XSS fuzzing against operational API boundaries.

## A04 Insecure Design

Insecure Design — PARTIALLY EVIDENCED: Order state transition guards (orders-transition.e2e-spec.ts), meal ordering window enforcement (ordering-window.e2e-spec.ts), allergy safety persistence/visibility (allergy-safety.e2e-spec.ts), and positive quantity/price domain rules are verified across 234 backend unit tests, 58 E2E tests, and 373 frontend Jest tests.

Remaining verification: Verify live cross-service FOSS-KMS-WKMS concurrency and physical KOT printer failover.

## A05 Security Misconfiguration

Security Misconfiguration — PARTIALLY EVIDENCED: Startup environment validation, disabled automatic schema synchronization, and managed PostgreSQL SSL checks pass in CI.

Remaining verification: Verify production CORS allowlists, error disclosure handling, and container hardening in staging.

## A06 Vulnerable and Outdated Components

Vulnerable and Outdated Components — PARTIALLY EVIDENCED (PROD AUDIT PASS): Both backend and frontend npm run security:audit steps passed with 0 production high/critical vulnerabilities in GitHub Actions Run 42502180570.

Remaining verification: Audit full development dependency trees before final release tagging.

## A07 Identification and Authentication Failures

Identification and Authentication Failures — PARTIALLY EVIDENCED: Frontend authentication state and kitchen role workflows are tested across 59 frontend Jest suites (373 passing tests) and API Gateway auth tests; live central-auth kitchen staff login E2E remains pending staging integration.

Remaining verification: Verify kitchen staff login, token expiry, session revocation, and lockout against live central auth.

## A08 Software and Data Integrity Failures

Software and Data Integrity Failures — PARTIALLY EVIDENCED: CI enforces npm ci with committed lockfiles, lint, build, and 665 automated tests in Run 42502180570.

Remaining verification: Verify signed RabbitMQ event envelopes and immutable container image provenance in staging.

## A09 Security Logging and Monitoring Failures

Security Logging and Monitoring Failures — PARTIALLY EVIDENCED: Order transition and kitchen operational state changes are tracked in application services; centralized SIEM log retention and security alerting remain unverified.

Remaining verification: Add dedicated security audit log redaction tests and staging monitoring alerts.

## A10 Server-Side Request Forgery

Server-Side Request Forgery — PARTIALLY EVIDENCED: KMS connects only to preconfigured PostgreSQL and RabbitMQ endpoints validated at startup; no user-controlled outbound URL fetching exists.

Remaining verification: Enforce container egress network policies in staging.

# 7. Worker Management System (WKMS)

Repository: forever-hotel/worker-management | Branch: develop | Tested revision: 1a33059c0005752ca709ca91ab11067dc6cff55a (PR #75 82f458005b; Oct 25 baseline: 0f91f1c9d6390c80c9f12f82b834fee9bf374104) | CI date: 9 November 2026

Evidence: https://github.com/forever-hotel/worker-management/actions/runs/42504698971, https://github.com/forever-hotel/worker-management/actions/runs/42504970394, https://github.com/forever-hotel/worker-management/actions/runs/42496517266

Dependency position: PASSING IN CI — Backend and frontend production high-threshold npm audits both passed with 0 vulnerabilities in Runs 42504698971 and 42504970394 alongside 649 passing automated tests (400 backend, 249 frontend) and >80% enforced whole-application branch and line coverage.

## A01 Broken Access Control

Broken Access Control — PARTIALLY EVIDENCED: Worker role JWT guards, task claim ownership checks, and receptionist manual assignment authorization (verified in WKMS unit suites and FDS-to-WKMS local W04-B verification: 6 PASS) are tested across 400 backend and 249 frontend tests on develop (1a33059c00). Evidence: backend/src/auth/guards/worker-role.guard.ts; backend/src/database/repositories/task.repository.spec.ts.

Required follow-up: Execute full cross-role E2E matrix on deployed staging.

## A02 Cryptographic Failures

Cryptographic Failures — PARTIALLY EVIDENCED: bcrypt password verification, signed JWT worker authentication, and Socket.IO token expiry enforcement (wkms-realtime.gateway.spec.ts, 94.91% line / 90.00% branch coverage) are verified in CI. Evidence: backend/src/auth/auth.service.spec.ts; backend/src/realtime/wkms-realtime.gateway.spec.ts.

Required follow-up: Verify staging HTTPS/WSS TLS certificates and secret rotation.

## A03 Injection

Injection — PARTIALLY EVIDENCED: Parameterized pg SQL queries and strict RabbitMQ message contract validators (checkout-completed.validator.spec.ts, order-ready.validator.spec.ts) reject malformed payloads. Evidence: backend/src/database/repositories/task.repository.ts; backend/src/messaging/contracts/checkout-completed.validator.spec.ts.

Required follow-up: Execute malicious SQL and AMQP payload fuzzing in staging.

## A04 Insecure Design

Insecure Design — PARTIALLY EVIDENCED: Row-level locking for the max-three-claimed-tasks rule, RabbitMQ message deduplication, and transactional escalation outbox dispatching (verified in unit tests and FDS Phase B local boundary run: 3 PASS) prevent double-claiming and lost escalations. Evidence: backend/src/database/repositories/task.repository.spec.ts; backend/src/messaging/outbox/outbox-dispatcher.service.spec.ts.

Required follow-up: Complete formal threat-model sign-off and live 15-minute escalation timer staging test.

## A05 Security Misconfiguration

Security Misconfiguration — PARTIALLY EVIDENCED: Joi environment validation, Socket.IO origin restrictions, and frontend public build configuration validation (enforced in CI workflow) prevent unsafe defaults. Evidence: backend/src/config/env.validation.spec.ts; backend/src/realtime/wkms-realtime.gateway.ts; .github/workflows/ci.yml.

Required follow-up: Configure production origin allowlists and inject release public URLs in deployment secrets.

## A06 Vulnerable and Outdated Components

Vulnerable and Outdated Components — PARTIALLY EVIDENCED (PROD AUDIT PASS): Backend and frontend production dependency audits passed with 0 vulnerabilities in Runs 42504698971 and 42504970394.

Required follow-up: Archive full-tree npm audit JSON alongside tagged release artifacts.

## A07 Identification and Authentication Failures

Identification and Authentication Failures — PARTIALLY EVIDENCED: Inactive worker rejection, non-WORKER role denial, JWT validation, and Socket.IO handshake authentication are tested in CI. Evidence: backend/src/auth/auth.service.spec.ts; backend/src/realtime/wkms-realtime.gateway.spec.ts.

Required follow-up: Verify login rate-limiting/lockout and central auth compatibility on staging.

## A08 Software and Data Integrity Failures

Software and Data Integrity Failures — PARTIALLY EVIDENCED: CI enforces npm ci, >=80% whole-app branch/line coverage gates, Docker image build verification (Run 42504970394), event schema version validation, and outbox retry persistence (verified across broker restart in FDS Phase C: 5 PASS). Evidence: backend/src/messaging/contracts/order-ready.validator.spec.ts; .github/workflows/ci.yml.

Required follow-up: Configure release environment URLs for the immutable image publishing job and verify image signing.

## A09 Security Logging and Monitoring Failures

Security Logging and Monitoring Failures — PARTIALLY EVIDENCED: Task lifecycle transitions, manual assignments, and escalation events are logged in service and outbox layers. Evidence: backend/src/tasks/task.service.ts; backend/src/tasks/task-escalation.service.spec.ts.

Required follow-up: Connect WKMS logs to centralized SIEM retention and incident alerting.

## A10 Server-Side Request Forgery

Server-Side Request Forgery — PARTIALLY EVIDENCED: Database and RabbitMQ connections use server-configured environment URLs only; WKMS does not fetch user-supplied URLs. Evidence: backend/src/config/env.validation.ts; backend/src/messaging/rabbitmq/rabbitmq.config.ts.

Required follow-up: Verify container network egress restrictions in staging.

# 8. Outstanding security acceptance

- Upgrade the remaining full-tree dependencies reported in the Hotel Website raw audit (Next.js 16.4.3 and NestJS/Express transitive overrides matching Manager Dashboard) and archive clean npm audit --json outputs for all repositories at the final release tag.
- Execute an integrated staging authorization and IDOR penetration test across forever-hotel/api-gateway, forever-hotel/websocket-service, and all six subsystem backends covering guest, worker, chef, receptionist, and manager roles.
- Verify deployed staging HTTPS/WSS TLS certificates, HSTS/CSP response headers, CORS origin allowlists, RabbitMQ broker TLS/vhost isolation, and container egress policies (blocking cloud metadata IP 169.254.169.254).
- Complete architectural sign-off and staging verification for the FOSS PostgreSQL transactional outbox (CR-outbox-foss.md) and Neon TypeORM migration history reconciliation in FDS (DDP-195).
- Obtain dated security reviewer sign-off and residual-risk disposition before certifying production OWASP compliance.